

Jiarui (Jerry) Yan

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EDUCATION

Carnegie Mellon University, School of Computer Science

Pittsburgh, PA

Master of Science in Computational Data Science GPA: 3.8/4.0

Expected 12/2026

Coursework: LLMs, Advanced NLP, Multimodal ML, DL, Cloud Computing, Distributed Systems, Computer Networks

University of Washington, Information School

Seattle, WA

Bachelor of Science in Informatics: Data Science Major, Applied Mathematics Minor

09/2021 – 06/2025

GPA: 3.94/4.0 | Dean's List (all terms) | Magna Cum Laude (top 3.5%)

Coursework: Machine Learning, NLP, Numerical Analysis, Probability I & II, Linear Algebra, Database Systems

RESEARCH INTEREST

LLM pre-training data, post-training, reinforcement learning, agentic systems, agent planning

PUBLICATIONS (FIRST / CO-FIRST AUTHOR)

[1] **J. Yan***, W. Sun*, S. Li, W. Li, Y. Yang. "TraceML: What Auto-Research Agents Miss in Long-Horizon ML Development." **NeurIPS 2026** (Evaluations & Datasets Track); also at the COLM 2026 WAB and NeurIPS 2026 IAE workshops.

[2] Z. Yu*, **J. Yan***, S. Sanghvi, N. Atri, C. Xiong. "ReScraper: Unified Scraping and Cleaning of Web Data for Effective LLM Pretraining." Under review.

[3] A. Kumar*, Z. Lei*, **J. Yan***, J. W. Momo, L. Reddy, R. E. Cabrera Jimenez, C. A. Cohen, A. Kajiyama, W. W. Cohen. "Learning to Construct Practical Agentic Systems." Under review.

[4] **J. Yan**, J. Tan, Y. Tan. "Will the Fork Be with You? How Forks Drive User Contributions on Open-Source Platforms." 35th POMS Conference (UTD24), ISOM Track, 2025.

*Equal contribution.

PUBLICATIONS (CONTRIBUTING AUTHOR)

[5] V. Madan, B. Miranda, Z. Zhou, S. Daruru, **J. Yan**, E. Obbad, S. Koyejo. "Which Environments Should LLMs Learn From? Predicting RL Utility Across Scale." Under review.

[6] Z. Lei, **J. Yan**, J. W. Momo, W. W. Cohen. "Learning Reusable Agent Subroutines from Experience." Under review.

RESEARCH EXPERIENCE (PROJECT LEAD)

TraceML: Long-Horizon Auto-Research Agents (NeurIPS 2026)

Pittsburgh, PA

Graduate Researcher, Prof. Yiming Yang (CMU LTI)

01/2026 – Present

- Built TraceML to capture how ML solutions are developed, not just where they end up: every scored code version of 4,465 human and 207 agent Kaggle trajectories, with two small labelers tagging what each edit did and why.
- Found that agents loop where human experts explore: Codex keeps re-weighting one ensemble, MLEvolve keeps mutating one model, only human pivots pay off, and top humans reopen dropped ideas on 9% of versions, Codex once in 658.
- Wrote a ~1K-token planning prompt from these gaps that lifted scores on 5 of 7 competitions with no regressions; released the data and an open-source toolkit that turns any CLI agent run into a report against the human cohorts.

ReScraper: Model-Based Pre-training Data Curation (under review)

Pittsburgh, PA

Graduate Researcher, Prof. Chenyan Xiong (CMU LTI)

01/2026 – Present

- Co-led ReScraper, one 0.6B model (Qwen3-0.6B, 32K context) that replaces the heuristic HTML scraper and dozens of rule-based filters: it extracts the main content of a page, then keeps, edits, deletes, or rewrites it in a single pass.
- Built the two-stage SFT pipeline on labels from three teachers (extraction, refining, rewriting) and ran vLLM inference over an 18M-page Common Crawl pool, at about a third of the extraction teacher's GPU hours.
- Pretrained 400M / 1.4B / 2.8B models on 8.2B / 28.8B / 55.9B tokens: best DCLM Core at every scale, +3.8–4.7% relative over the strongest baseline; open-sourced the code, model, and 7.44B-token corpus.

RESEARCH EXPERIENCE (COLLABORATIONS)

Predicting RL Utility Across Scale (under review)

Palo Alto, CA

Graduate Researcher, Prof. Sanmi Koyejo (Stanford SAIL)

05/2026 – 09/2026

- Helped predict how much RL post-training improves a model from measurable environment properties, across model scale.
- Designed metrics that make these properties controllable and cheap to measure, plus controlled experiments holding the RL algorithm fixed to isolate environment effects across math, coding, and tool-use tasks.

Agentic LLM Systems with Pseudo-Tools

Pittsburgh, PA

Graduate Researcher, Prof. William W. Cohen (CMU MLD, Google DeepMind)

01/2026 – 05/2026

- Built a modular agentic-LLM Python framework on Pydantic-AI with LiteLLM routing across 100+ model APIs, exposing typed interfaces swappable between Python functions, LLM calls, ReAct sub-agents, and sandboxed code-gen backends.
- Developed three learning algorithms (pseudo-tool inducer, code distillation, orchestration hill-climbing) and integrated NSGA-II evolutionary search over YAML-encoded configs, reducing sweep cost up to 8x via LLM caching.
- Benchmarked 19 evaluation suites (BBH, FinQA, τ -Bench, RuleArena) on DeepSeek-V3.1; static workflows hit 0.64 avg correctness vs 0.56 ReAct at half the cost, and the learned orchestrator reached 0.76 vs 0.55 human-engineered baseline.

PRIOR RESEARCH EXPERIENCE

Transfer Learning for Recommendation Systems

Seattle, WA

Research Assistant, Prof. Wanning Chen (UW, Foster School)

11/2023 – 12/2024

- Deployed a transfer-learning matrix completion framework using proxy representations and SVD low-rank approximation to improve recommendation prediction on sparse target domains.
- Aggregated 2M+ NetEase users and content into 200×70 interaction matrices; tuned hyperparameters via cross-validation on Frobenius norms, reducing RMSE from 0.6 to 0.35.

Retrieval-Augmented Generation (RAG) Question Answering System

Seattle, WA

Student Researcher, Prof. Ranjay Krishna (UW, CSE)

01/2024 – 06/2024

- Built a multimodal RAG system over course materials: extracted 2M+ tokens via PyDub, VOSK speech-to-text, and PyTesseract OCR, embedding documents with BERT for semantic retrieval.
- Integrated LLaMA-2 (13B) with a Pinecone vector database, achieving 82% precision and 77% recall on 200+ evaluation queries; served 50+ concurrent users on Lightning AI with GPU acceleration.

INDUSTRY EXPERIENCE

Google LLC

Sunnyvale, CA

Software Engineer Intern

05/2026 – 09/2026

- Built an LLM agent for non-unilateral access control automating Google Workspace SRE access authorization, parsing natural-language peer-approval signals in Buganizer tickets into structured, policy-compliant endorsements.
- Hardened the agent with deterministic guardrails, metadata-grounded requester/approver checks, prompt-injection risk scoring, and immutable-history validation, defending against injection attacks and related risks.
- Built the evaluation stack: a golden dataset of human-annotated and synthetic adversarial cases with daily automated monitoring, reaching 93.75% F1; shipped 23 CLs across API, agent, and eval infrastructure.

Quantum Crypto

Beijing, China

Research Intern

01/2025 – 08/2025

- Designed an end-to-end data processing system with Apache Spark and Hadoop to handle 1TB+ daily financial data ingestion and feature generation.
- Developed an LLM-based Q&A and fund-recommendation agent on Qwen2.5-14B with a full RAG pipeline (query rewrite, retrieval, reranking, prompt engineering); embedded FAQ and fund-profile corpora with sentence-transformers (all-MiniLM), reducing routine support workload by 25% while increasing purchase inquiries.
- Built a transaction-monitoring backend with FastAPI and PostgreSQL; instrumented system metrics and Grafana dashboards for real-time account observability.

Elile AI

Seattle, WA

Machine Learning Engineer Intern

05/2024 – 08/2024

- Engineered an automated ETL pipeline using Airflow, processing 1M+ sensor records into structured datasets in AWS RDS.
- Trained XGBoost predictive-maintenance models for 4K+ EV charging stations, achieving 87% accuracy predicting failures 7 days in advance and improving uptime by 12%.
- Containerized ML services via Docker with CI/CD pipelines, reducing deployment time from 2 hours to 30 minutes (75%).

Huawei Technologies

Beijing, China

Data & Algorithm Engineer Intern

06/2023 – 09/2023

- Architected a distributed pipeline for 10M+ geographical Point-of-Interest entries (Apache Spark, Hadoop HDFS, Elasticsearch), saving 90% of manual classification time.
- Enhanced geospatial aggregation with Area-of-Interest feature engineering in Java and real-time ingestion via Kafka streaming, improving clustering accuracy by 8%.
- Fine-tuned BERT and word-embedding models (PyTorch, NLTK) for duplicate detection, achieving 0.89 precision and 0.76 F1 on 100K validation entries.

TEACHING

11-631: Data Science Seminar (Fall 2026) Prof. Maarten Sap

- Teaching assistant for the required MCDS core seminar; support weekly sessions, grade project deliverables, and advise student teams on scoping and presenting data science research.

INFO 498: Text Mining & Analytics (Spring 2023, Fall 2023) Prof. Lucy Lu Wang

INFO 330: Databases & Data Modeling (Spring 2024) Prof. Lucy Lu Wang

- Designed assignments and evaluated projects covering corpus analysis, TF-IDF retrieval, text classification, and information extraction; supported students on NLTK / scikit-learn / spaCy pipelines.
- Led weekly SQL recitations and group project development for 35 students; gave individualized feedback on relational data modeling and query design.

SKILLS

Programming Languages: Python, C++, Java, TypeScript, JavaScript, SQL, R, MATLAB, Bash, Shell

ML & LLM: PyTorch, vLLM, Verl, Hugging Face, TensorFlow, Scikit-learn, Pydantic-AI, LiteLLM, LangChain, RAG, CUDA

Data & Distributed Systems: Apache Spark, Hadoop, HDFS, Hive, Kafka, Airflow, ETL Pipelines, Parquet, NumPy, Pandas

Cloud & DevOps: AWS, Google Cloud Platform, Azure, Docker, Kubernetes, Git, CI/CD, Terraform, Linux

Databases & Search: PostgreSQL, MySQL, Redis, MongoDB, Pinecone, Elasticsearch